



IISER Mohali
[August 2025]
PHY424 (QFT-I)

Mid-Sem I

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Max. Marks : 20

- ✓ 1. In the context of quantization of a non-interacting complex scalar field,
 - (a) Construct the Fock space by specifying vacuum state, one-particle states, two-particle states and a general multi-particle state.
 - (b) Calculate $Q(a_p^\dagger b_p^\dagger |0\rangle)$ explicitly, where Q is the charge operator.

[2]
- ? 2. Show that for time-like separations, $\text{sign}(x^0 - y^0)$ is invariant under Lorentz boost. Further, argue that the definition of time ordered product $T\phi(x)\phi^\dagger(y)$ is Lorentz invariant.

[4]
3. Show that the mass operator in complex scalar field theory, $\mu(x) = \phi^\dagger(x)\phi(x)$ satisfies causality.

[2]
- ✓ 4. Consider two complex scalar fields for which,
$$\mathcal{L} = \partial_\mu \phi_a^* \partial^\mu \phi_a - m^2 \phi_a^* \phi_a, \quad a = 1, 2$$
 - (a) Show that $\phi_a \rightarrow U_{ab} \phi_b$ where $U = e^{i\alpha \cdot \sigma}$ is a symmetry of the above Lagrangian. Here σ^k are Pauli matrices.
 - (b) Derive the conserved charges in terms of fields and their conjugate momenta.

[4]
- ✓ 5. Show that for a massless Dirac particle, $u(p)$ is an eigenstate of $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$.

[2]
- ✓ 6. Derive the Lorentz transformation property of $\bar{\psi}\gamma^\mu\partial_\mu\psi$.

[2]
7. Derive the energy-momentum tensor $T^{\mu\nu}$ for,

$$\mathcal{L} = \partial^\mu \phi \partial_\mu \phi - g x^\mu x_\mu \phi^2$$

where, g is a constant. Further, find the divergence $\partial_\mu T^{\mu\nu}$, and interpret the result.

[4]

$$g \frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{\partial \mathcal{L}}{\partial x^\mu} = 2g x_\mu \phi^2$$